TITLE PAGE

- **Problem Statement ID – 1**
- **Problem Statement Title- Eco-Loop Building Agents**
- **Theme- Smart Automation & Sustainability**
- **PS Category- Software**
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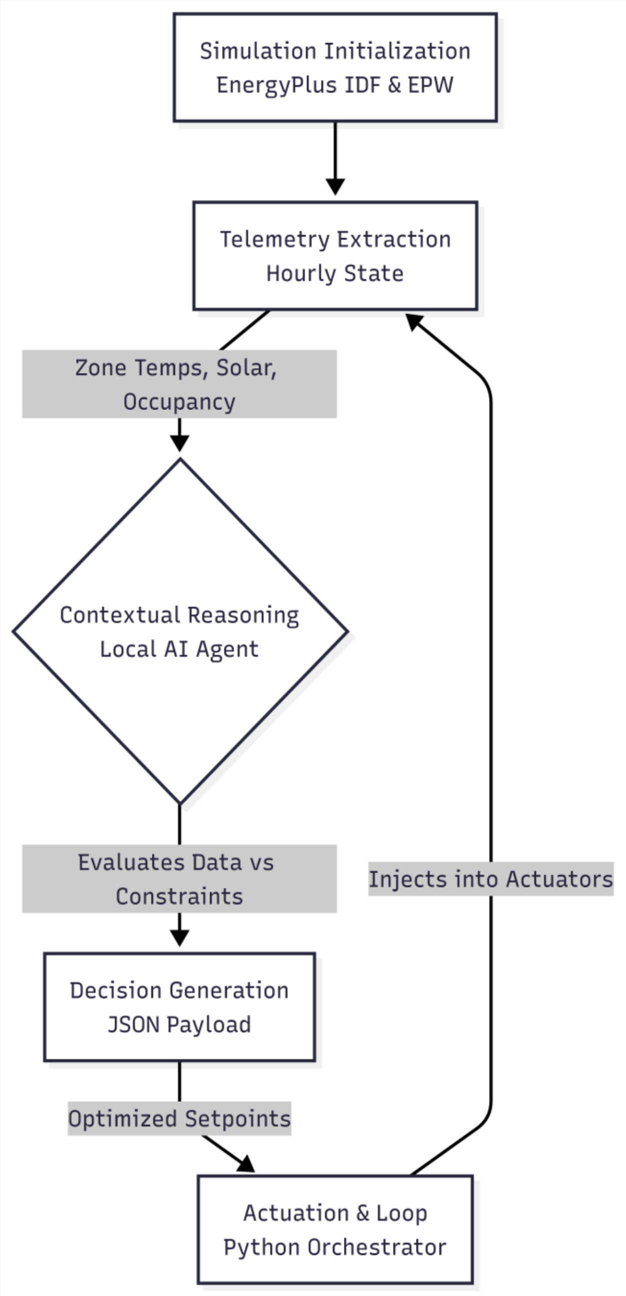
Eco-Loop AI: Autonomous Multi-Zone HVAC & Carbon-Aware Building Orchestration

Proposed Solution

- Eco-Loop AI is a closed-loop agentic orchestrator integrating EnergyPlus (DOE simulation engine) with a local LLM (Qwen2.5:14b via Ollama) to optimize 7-zone HVAC setpoints in real time.
- Addresses the problem: Buildings consume ~40% of global energy. Traditional BMS use rigid static schedules, wasting energy during unoccupied hours & ignoring dynamic grid carbon conditions.
- Innovation & Uniqueness: (1) Predictive Schedule Lookup – anticipates thermal loads via next_hour_occupancy. (2) 1.0° C/hr Anticipatory Pre-Cooling Ramp – prevents grid shock. (3) Carbon-Aware Intelligence – coasts setpoints during high-carbon grid hours. (4) Fully Local Edge AI via Ollama – zero cloud dependency, 6 GB VRAM device.



TECHNICAL APPROACH

- <u>Simulation Engine:</u>

• EnergyPlus v22.1.0 (DOE) via pyenergyplus API | AI Engine: Qwen2.5:14b running locally via Ollama (open-weights, edge device) | Language: Python 3.11 | Libraries: pyenergyplus, Pandas, asyncio, re (regex) | Hardware: 6 GB VRAM edge device with CPU/GPU partial offloading

- <u>Implementation:</u>

• (1) Python Orchestrator binds to EnergyPlus API & hooks callback.

• (2) Each simulated hour: extract zone temps, solar, occupancy, grid carbon intensity.

• (3) Package telemetry as JSON prompt → send to local LLM.

• (4) Defensive parsing pipeline: strip markdown blocks → strip // inline comments → json.loads() → fallback regex on exception.

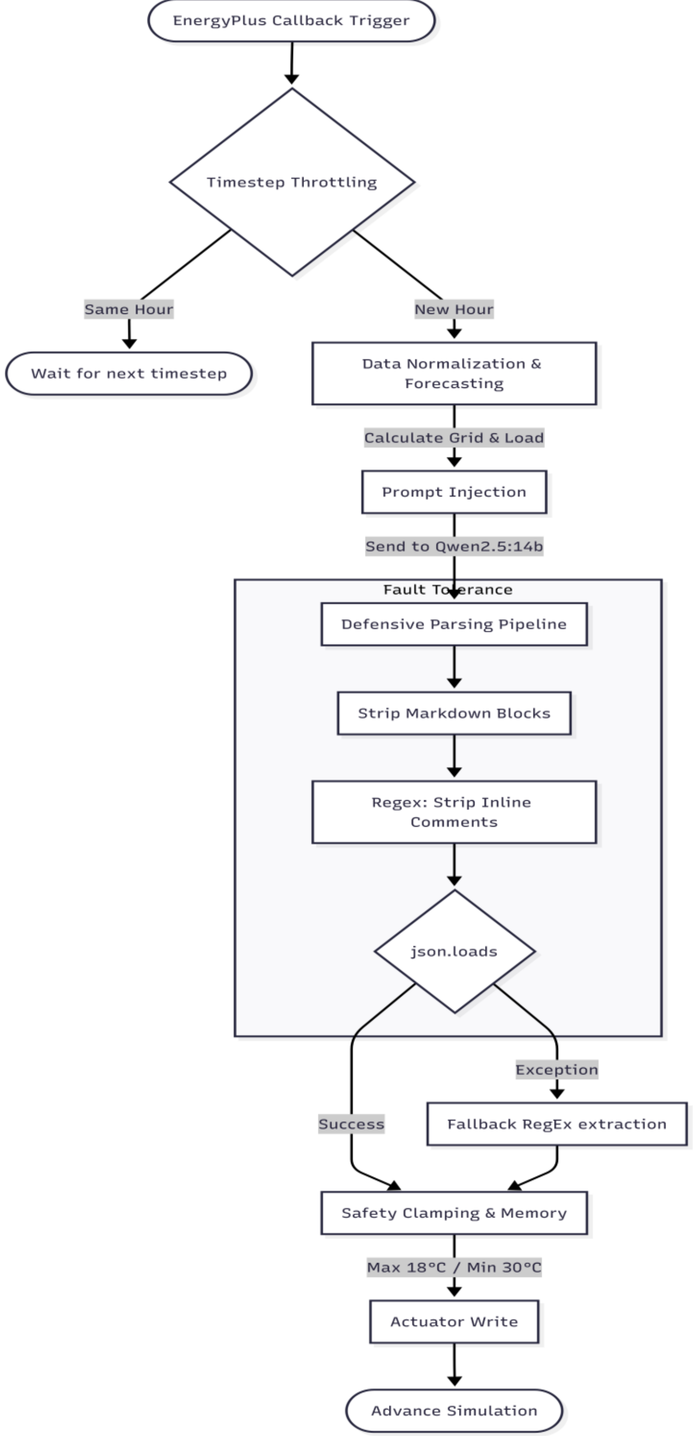
• (5) Safety-clamp setpoints to 18° C–30° C → inject into EnergyPlus actuators. High-level & Low-level architecture diagrams are embedded below.

FEASIBILITY AND VIABILITY

- **Feasibility:** Successfully tested on a 7-day continuous EnergyPlus simulation using DOE Commercial Prototype Medium Office model, Climate Zone 2A (Tampa, FL). Achieved ~1% total energy reduction vs rule-based baseline while strictly enforcing occupant comfort bands (20° C–26° C). 100% uptime across thousands of simulated hours.
- **Challenges:** (1) LLM Syntax Variance – local models hallucinate markdown/comments that crash standard JSON parsers. (2) Hardware Bottleneck – 14B-parameter model on 6 GB VRAM; per-15-min inference is unfeasible. (3) Thermal Inertia Lag – rapid morning warm-up if HVAC is set back too aggressively overnight.
- **Strategies to overcome:** (1) Engineered custom multi-stage regex pipeline (re.sub(r'//.*','')) to strip all LLM filler before json.loads(); fallback key-value regex on exception – guarantees crash-free operation. (2) Hourly timestep orchestration: callback fires every EnergyPlus timestep but AI is only invoked once per simulated hour, keeping latency manageable. (3) Implemented 1.0° C/hr pre-cooling ramp via last_commanded_setpoint memory – zones hit 24° C targets precisely as

ARTIFACTS

- Relevant artifacts (to be embedded / linked in final submission):
  - Source Code: orchestrator.py – EnergyPlus API callback loop & defensive JSON parsing pipeline (GitHub repository link to be added)
  - Solution Architecture Diagrams: High-level Feedback-Reasoning-Control loop & Low-level EnergyPlus callback workflow (see Technical Approach slide)
  - Dashboard Screenshots: Streamlit UI – Normalized kWh Analytics panel & Small-Multiples Thermal Tracking vs Baseline (to be added manually)



RESEARCH AND REFERENCES

Details / Links of the reference and research work

- DOE/PNNL Commercial Prototype Building Models – Medium Office, Climate Zone 2A: https://www.energycodes.gov/prototype-building-models
- EnergyPlus™ v22.1.0 Documentation – U.S. Department of Energy: https://energyplus.net
- Ollama – Local LLM inference framework (Qwen2.5:14b): https://ollama.com
- pyenergyplus – Python API wrapper for EnergyPlus runtime: https://github.com/NREL/EnergyPlus/tree/develop/pyenergyplus
- Weather Data: USA_FL_Tampa-MacDill.AFB.722110_TMY3.epw – EnergyPlus Weather Files